

Week 5 Technical Documentation: Data Preprocessing, Pipelines & Unsupervised Clustering

1. Executive Summary & Week 5 Objectives

This report documents the completion of Week 5 milestones for the Non-Mortgage Living Expense Estimation Engine covering Harris, Fort Bend, and Galveston counties (279 canonical ZIP codes). During this phase, all multi-domain data sources (demographic crosswalks, NFIP claims, FEMA spatial flood hazards, utility rate schedules, and USDA food baselines) were audited, cleaned, engineered, transformed into machine-readable feature stores, and clustered into regional cost and risk archetypes.

The deliverables fulfilled include:

- **Data Cleaning & Quality Assessment:** Enforced zero missing values across the 279 ZIP code index, preserved string formatting for primary keys, and filtered out non-differentiating baseline constants.
- **Feature Engineering:** Constructed domain-specific spatial hazard indicators, utility rate metrics, municipal utility district (MUD) flags, and statutory windstorm insurance requirements.
- **Data Transformation Pipelines:** Designed and executed a Scikit-Learn ColumnTransformer pipeline incorporating StandardScaler and OneHotEncoder to prepare an unskewed matrix for machine learning.
- **Train / Validation / Test Splitting Adaptation:** Adapted traditional holdout splits to an unsupervised clustering framework across the complete geographic universe, utilizing the Elbow Method, Silhouette Analysis, and the Davies-Bouldin Index for validation.
- **Model-Ready Artifacts:** Successfully generated and validated `features_scaled_for_clustering.parquet` and `master_living_costs_clustered.parquet`.

2. Data Cleaning & Quality Assessment

A rigorous quality audit was performed on the consolidated dataset before pipeline transformation:

- **Primary Key String Integrity:** Applied zero-padding (`str.zfill(5)`) to ensure all USPS ZIP codes maintain exact 5-character string lengths (preventing numerical truncation such as 07702 converting to 7702).
- **Record Deduplication:** Verified exactly 279 unique ZIP codes matching the HUD-USPS

residential ratio allocation (Harris: 224, Fort Bend: 31, Galveston: 24).

- **Null Value Remediation:** Zero null or missing values exist across all 29 original columns. Missing claims in inland ZIP codes were cleanly mapped to 0.0 to preserve analytical continuity.
- **Zero-Variance Column Pruning:** Identified and isolated uniform static variables (such as gas_base_monthly_fee at \$18.50, houston_cpi_food_scalar at 1.045, and USDA baseline tiers) to prevent zero-variance columns from artificially weighting distance calculations.

3. Feature Engineering Architecture

To balance physical hazards against regulated infrastructure pricing, 20 numerical and encoded features were engineered for the clustering engine:

Feature Category	Feature Name	Data Type	Transformation Applied	Domain Logic & Source
Flood & Spatial Hazards	pct_area_100yr_flood	float64	StandardScaler	Area percentage within 100-year AE floodplain.
	pct_area_coastal_surge	float64	StandardScaler	Area percentage within high-velocity VE coastal surge zone.
	est_flood_risk_tier	Categorical	OneHotEncoder (3 cols)	Categorical tier (VE: Coastal, AE: Floodplain, X: Minimal).
Insurance History	nfip_total_claims	float64	StandardScaler	Historical flood claim frequency per ZIP code.
	nfip_avg_building_claim	float64	StandardScaler	Mean historical structural damage claim payout (\$/claim).
	nfip_avg_contents_claim	float64	StandardScaler	Mean personal property contents claim payout (\$/claim).
Utility Delivery Tariffs	elec_base_monthly_fee	float64	StandardScaler	Fixed electric customer

Feature Category	Feature Name	Data Type	Transformation Applied	Domain Logic & Source
				connection fee (\$4.39 vs \$7.85).
	elec_delivery_rate_per_kwh	float64	StandardScaler	Regulated electric volumetric delivery tariff (\$/kWh).
	elec_provider	Categorical	OneHotEncoder (2 cols)	Utility territory (CenterPoint vs. TNMP).
	gas_distribution_rate_per_ccf	float64	StandardScaler	Natural gas distribution tariff (\$0.0912 vs \$0.0985/Ccf).
Municipal Water & MUDs	water_base_monthly_fee	float64	StandardScaler	Fixed water base charge (\$16.80 City vs \$28.50 MUD).
	water_rate_per_1k_gal	float64	StandardScaler	Volumetric rate per 1,000 gallons (\$6.45 City vs \$8.50 MUD).
	water_service_type	Categorical	OneHotEncoder (2 cols)	Municipal_CityOfHouston vs. Suburban_MUD.
Statutory Indicators	water_mud_flag	int64	Passthrough (Binary)	1 if inside a Municipal Utility District; 0 if Municipal.
	twia_mandatory	int64	Passthrough (Binary)	1 if in TWIA catastrophe zone (Galveston & Coastal Harris).

4. Data Transformation Pipeline

Because unsupervised K-Means clustering depends directly on Euclidean distance calculations, large unscaled numerical ranges (such as flood percentages from 0 to 100% versus electric tariffs from \$0.0381 to \$7.85) would disproportionately distort cluster boundaries. A Scikit-Learn ColumnTransformer pipeline was implemented to scale continuous variables and encode categorical attributes:

```
preprocessor = ColumnTransformer(  
    transformers=[  
        ('num', StandardScaler(), numeric_cols),  
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False),  
categorical_cols),  
        ('pass', 'passthrough', passthrough_binary_cols)  
    ],  
    remainder='drop' # Explicitly eliminates text strings and static baseline  
constants  
)
```

The pipeline exported features_scaled_for_clustering.parquet, containing 279 rows, 1 primary key string (zip_code), and 20 float64 mathematical dimensions.

5. Train / Validation Splitting Strategy & Cluster Optimization

A standard 70/15/15 train/test split was adapted for this capstone to align with the hybrid architecture. Because the goal is to partition the entire contiguous metropolitan area of 279 ZIP codes into macro-archetypes rather than predicting a synthetic supervised total, the entire scaled feature matrix was evaluated using internal validation metrics:

- **Elbow Method (Inertia / WCSS):** Quantified intra-cluster compactness across cluster counts $k = 2$ through $k = 6$.
- **Silhouette Analysis:** Evaluated cluster cohesion and separation distance (higher values represent superior separation).
- **Davies-Bouldin Index:** Measured the similarity between each cluster and its most similar counterpart (lower values indicate cleaner partitioning).

Number of Clusters (k)	Silhouette Score (Higher is Better)	Davies-Bouldin Index (Lower is Better)	Inertia (WCSS)	Model Assessment
k = 2	0.5954	0.9781	1858.94	Underfit; broad

Number of Clusters (k)	Silhouette Score (Higher is Better)	Davies-Bouldin Index (Lower is Better)	Inertia (WCSS)	Model Assessment
				county-level split lacking infrastructure nuance.
k = 3	0.6375	0.8112	1208.77	Moderate; captures basic MUD vs. City boundary but misses flood tiers.
k = 4	0.6670	0.6986	911.58	OPTIMAL DELIVERABLE: High separation; balanced, robust cluster sizes.
k = 5	0.6698	0.5679	634.73	Mathematical peak, but suffered from cluster degeneracy (singleton n=1).
k = 6	0.5310	0.7365	516.18	Overfit; sharp drop in silhouette score indicating fragmented clusters.

6. Technical Challenges, Errors & Resolution Log

Throughout Week 5 execution, four major technical challenges were encountered and successfully resolved:

Challenge 1: Google Drive Path Mismatch and Virtual Filesystem Cache Lag

Issue: Execution in Colab threw FileNotFoundError: [Errno 2] No such file or directory when loading the master dataset, despite files being visible in Google Drive.

Root Cause: Two compounding factors: (1) A minor naming discrepancy in the root folder containing a hyphen (AI Resource Capstone - 2277 vs AI Resource Capstone 2277), and (2) Google Colab's background virtual drive cache failing to index files moved in the web UI without a forced remount.

Solution: Executed an explicit Drive search routine, locked in the verified hyphenated directory string, and passed force_remount=True to synchronize the filesystem.

Challenge 2: Scikit-Learn Vectorization Crash (String Blocker)

Issue: The pipeline passed remainder__gas_provider ('CenterPoint') through to the clustering matrix, which would trigger ValueError: could not convert string to float during distance computations.

Root Cause: The ColumnTransformer previously utilized remainder='passthrough' without explicitly dropping non-encoded string columns.

Solution: Redesigned the preprocessing pipeline to explicitly select numeric and one-hot encoded columns while specifying remainder='drop', ensuring 100% float64 type conformity.

Challenge 3: Zero-Variance Feature Distortions

Issue: Columns including gas_base_monthly_fee (\$18.50), houston_cpi_food_scalar (1.045), and the USDA baseline food tiers held constant values across all 279 rows.

Root Cause: Inclusion of static constants adds zero variance to distance computations and unnecessarily increases dimensionality.

Solution: Filtered out static cost metrics from the clustering feature set while retaining them in the master dataset for downstream deterministic cost evaluation in the Streamlit web engine.

Challenge 4: Cluster Degeneracy & Singleton Anomaly Identification

Issue: While $k = 5$ achieved the highest raw mathematical Silhouette Score (0.6698), empirical inspection of cluster counts revealed: Cluster 0: 193, Cluster 1: 1, Cluster 2: 13, Cluster 3: 58, and Cluster 4: 14.

Root Cause: Cluster 1 isolated a single ZIP code (72225, an administrative Houston PO Box / Medical Center zone). Presenting an archetype with $n=1$ in a project defense represents an overfit anomaly rather than an actionable macro-cluster.

Solution: Selected $k = 4$ as the production configuration. As verified by the metrics, $k = 4$ maintains a strong Silhouette Score of 0.6670 while producing well-distributed, defensible groups (193, 59, 14, and 13 ZIPs) without singletons.

7. Final Cluster Archetypes & Deliverables ($k = 4$)

The 279 ZIP codes were partitioned into four distinct regional archetypes and exported to master_living_costs_clustered.parquet and master_living_costs_clustered.csv:

Cluster ID	Archetype Designation	ZIP Count	Infrastructure & Tariff Profile	Environmental & Hazard Risk Profile
Cluster 0	Urban Core / Municipal Stable	193 ZIPs	City of Houston Water (\$16.80 base, \$6.45/1k gal), CenterPoint Electric (\$4.39 base, \$0.0381/kWh).	Low flood exposure (1.73% 100-yr flood, 0.0% surge). Zero TWIA mandatory insurance requirements.
Cluster 1	Galveston Mainland / Coastal Utility Corridor	14 ZIPs	TNMP Electric Service Territory (\$7.85 base fee, \$0.0462/kWh delivery), MUD water governance.	Mandatory TWIA windstorm catastrophe coverage (1.0). Moderate inland drainage (5.00% 100-yr flood).
Cluster 2	Suburban MUD Growth Corridor	59 ZIPs	Suburban MUD water districts (\$28.50 base, \$8.50/1k gal), CenterPoint Electric (\$4.39 base).	Inland suburban drainage (4.08% 100-yr flood, 0.0% surge). Zero TWIA mandatory requirements.
Cluster 3	High-Hazard Coastal Surge & Floodplain	13 ZIPs	High blended utility delivery rates, mixed MUD/municipal service territories.	Critical coastal surge exposure (18.5% surge coverage, 42.0% 100-yr floodplain). 100% mandatory TWIA coverage.

8. Class Activities & Weekly Progress Log

- **Team Working Sessions:** Executed end-to-end integration across all intermediate data sections, standardizing primary key joins across 279 ZIP codes. Aligned project

architecture with documentation in Google Docs.

- **Technical Problem-Solving:** Overcame Google Colab cache inconsistencies, eliminated data type crashes by sanitizing raw string columns, and diagnosed cluster degeneracy to justify $k = 4$ over $k = 5$.
- **Peer Assistance:** Demonstrated how to utilize Scikit-Learn ColumnTransformer objects to handle heterogeneous continuous, categorical, and binary data without leakage. Advised peers on chunking large multi-gigabyte datasets (such as NFIP claims) to bypass memory overflow crashes.